

Long, filmy hind wing strengthened by network of veins

Thick leading edge reinforces hind wing

Wing attached to central body section, or thorax, which contains wing muscles

Up and away

Some insects can take off in a split second, but a ladybug takes time to get airborne. Like most flying insects, it has two pairs of wings, but only its hind wings beat when it flies. Its forewings, or elytra, are thick and curved. Just before takeoff, the elytra swing outward, letting its delicate hind wings unfold. Once they start beating, the ladybug lets go with its legs and launches itself into the air.

Wing curves during flight to scoop flying insects into bat's mouth

Long finger bones stretch skin tight

Flying styles

Birds flutter or swoop overhead depending on their wing shape. A barn owl has broad wings, and it beats them slowly as it hunts animals on the ground. Swifts have narrow, sharply pointed wings with a backswept curve. This is an ideal shape for swerving at high speed, which lets swifts catch insects in midair. Woodpeckers have broad, tapering wings. They flap them in bursts, giving a flight path that rises and falls.

Barn owl flight path

Swift flight path

Woodpecker flight path

Skinny wings

There are more than a thousand kinds of bat, and all of them have wings made from a double layer of skin. In a bat's wing, the main framework is made up by the arm bones, while the skin is stretched tight by extremely long and slender fingers. Most bats have hooklike thumbs, and a separate flap of skin stretched between their legs. This long-eared bat has a wingspan of 12 in (30 cm), but the largest fruit-eating bats have wingspans of up to 5 ft (1.5 m).